

Gursimran Singh

Staff AI Researcher | LLM Systems, RL Fine-Tuning, and Multimodal AI

Vancouver, BC, Canada | gursimran.ubc@gmail.com | +1 604 614 9692

gursimar.github.io | github.com/gursimar | linkedin.com/in/gursimar

Technical Skills

Specializations: LLM Inference Optimization, Reinforcement Learning, Model Efficiency, Distributed Training/ Inference.

Programming: Python, C++, CUDA, Ascend-CANN

Frameworks and systems: PyTorch, vLLM, Megatron-LM, AReaL, Ray, Transformers, PyTorch3D.

Professional Experience

Huawei Canada Vancouver, BC, Canada

Staff AI Researcher

Senior AI Researcher

AI/ML Researcher

May 2019 – Present

Apr 2026 – Present

Nov 2023 – Apr 2026

May 2019 – Nov 2023

LLM/ LMM inference optimization, elastic serving, and reinforcement learning

- Identified and secured support for new research directions in LLM inference, elastic serving, and RL fine-tuning infrastructure.
- Led lean technical teams and served as a hands-on architect across system design, implementation, evaluation, and delivery.
- Project lead for research programs in multimodal-model inference, elastic MoE serving, and multi-tenant RL infrastructure.
- Took projects from problem formulation to top-tier publication, patenting, open-source release, and product integration.
- **Outcomes:** 8 projects delivered, 13 patent filings, 10 papers and manuscripts, multiple open-source contributions, and 7 awards.

Aspiring Minds (acquired by SHL) Gurugram, India

Senior Research Engineer

Research & Development Engineer

Applied machine-learning products, automated assessment, and team mentorship

Jul 2013 – Jun 2017

Jan 2017 – Jun 2017

Jul 2013 – Dec 2016

- Led research and development for an end-to-end ML product for automated programming assessment, from problem formulation and rubric design to data collection, modeling, evaluation, and deployment.
- Developed a question-independent, semi-supervised grading approach that reduced question-development effort by 5×.
- Built and maintained production scoring systems, mentored research engineers, and helped build and hire a 15-person team developing programming assessments and related products.
- **Outcomes:** 5× reduction in question-development effort, deployment at 1M+ programmer scale, adoption by customers including Amazon, and a KDD 2016 publication.

Other Roles

- Graduate Research Assistant — *University of British Columbia* May 2018 – Oct 2019
- Graduate Teaching Assistant, Master of Data Science program — *University of British Columbia* Sep 2017 – Apr 2019
- Research Intern — *Indian Institute of Science (IISc)* Jan 2012 – May 2012
- KVPY Research Intern — *Indian Institute of Technology Bombay (IIT)* Jun 2011 – Jul 2011

Selected Research Programs

Efficient Large Multimodal Model Inference | Project Lead / Core Contributor

ICML 2025; CVPR 2025

EPD Disaggregation: [Code]

- Led a five-person team and architected encode–prefill–decode disaggregation for independently scaling multimodal model-serving stages.
- Reduced memory use by up to 15×, supported 10× more images per request, and lowered TTFT by up to 71%.
- Published at ICML 2025, selected as a Huawei high-value patent, and adopted by vLLM, SGLang, NVIDIA Dynamo, and xLLM-service.

DivPrune: [Code]

- Co-led a three-person research effort covering prototyping, experimentation, and paper development for training-free, pre-attention visual-token pruning.
- Pruned up to 60% of visual tokens with negligible accuracy loss, enabled up to 10× more video frames, and reduced TTFT by roughly 50%.
- Published at CVPR 2025; received 200+ citations and community integrations for models including Gemma 4 and Qwen2.5-VL.

MARLaaS:

- Co-designed a multi-tenant asynchronous RL service that decouples rollout generation, environment interaction, and policy training.
- Enabled concurrent RL workloads to share distributed inference and training infrastructure, improving accelerator utilization and reducing duplicated hardware resources.
- Contributed to system architecture, implementation, large-scale evaluation, and development of the research manuscript.

LoRA-based RL Finetuning:

- Led the open-source integration of LoRA RL fine-tuning into AReal's Megatron engine using Megatron Bridge.
- Implemented model construction, distributed adapter training and export, and Megatron-to-vLLM adapter synchronization.
- Delivered GPU and Ascend support, including pipeline parallelism, single- and multi-node MoE training, and in-memory XCCL weight updates. [PR 1056] [PR 1123] [PR 1145] [PR 1159] [PR 679]

Selected Publications and Preprints

1. Gursimran Singh, X. Wang, I. Hu, T. Yu, L. Xing, W. Jiang, Z. Wang, X. Bai, Y. Li, et al. "Efficiently Serving Large Multimedia Models Using EPD Disaggregation". *ICML 2025*.
2. S. R. Alvar, Gursimran Singh, M. Akbari, and Y. Zhang. "DivPrune: Diversity-Based Visual Token Pruning for Large Multimodal Models". *CVPR 2025*.
3. Gursimran Singh, T. Yu, H. Li, C. Chen, H. Sadri, Q. Zhang, Y. Zhang, Y. Xiong, et al. "ElasticMoE: An Efficient Auto Scaling Method for Mixture-of-Experts Models". *Preprint, under review, 2025*.
4. T. T. L. Yu, Gursimran Singh, G. Shi, H. Sadri, Y. Zhang, and Z. Fan. "MARLaaS: Multi-Tenant Asynchronous Reinforcement Learning as a Service". *Preprint, under review, 2026*.
5. Gursimran Singh, T. Hu, M. Akbari, Q. Tang, and Y. Zhang. "Towards Secure and Usable 3D Assets: A Novel Framework for Automatic Visible Watermarking". *WACV 2025, Oral*.
6. M. L. Heisler, L. Xing, G. Shi, H. Sadri, Gursimran Singh, W. Zhang, T. Ye, Y. Xiong, et al. "Enhancing Learned Knowledge in LoRA Adapters Through Efficient Contrastive Decoding on Ascend NPUs". *KDD 2025*.
7. Gursimran Singh, L. Chu, L. Wang, J. Pei, Q. Tian, and Y. Zhang. "Mining Minority-Class Examples with Uncertainty Estimates". *MMM 2022, Oral*.
8. Gursimran Singh, L. Sigal, and J. J. Little. "Spatio-Temporal Relational Reasoning for Video Question Answering". *BMVC 2019, Oral*.
9. Gursimran Singh, S. Srikant, and V. Aggarwal. "Question Independent Grading Using Machine Learning: The Case of Computer Program Grading". *KDD 2016, Oral*.

Reviewing: Reviewer for *NeurIPS*, *CVPR*, *ECCV*, *AAAI*, *WACV*, *IEEE TSC*, and *IEEE ICASSP*; 16+ completed manuscript reviews.

Patents, Open-Source, and Awards**Patents (14 total)**

- First inventor on 8 U.S. patents, with 6 additional published patent applications (14 total).
- Patent portfolio spans machine-learning data mining, privacy, fairness, watermarking, model selection, and efficient AI systems.
- The patent on *EPD Disaggregation* invention received Huawei's High-Value Patent recognition.
- The patent on *Mining Minority-Class* has been cited by organizations including Microsoft, IBM, Intel, NVIDIA, and NEC.

Open Source (GitHub Contributions).

- Contributor to **AReal**; implemented Megatron Bridge integration, LoRA RL training, adapter export and synchronization, pipeline parallelism, multi-node MoE support, and GPU/Ascend execution paths. [PR 1056] [PR 1123] [PR 1145] [PR 1159] [PR 679]
- Additional open-source projects spanning **EPDServe**, **ElasticMoE** and **3D visual watermarking**.
- *Community*: Former coordinator for ML India, organizing ML meetups, technical talks, workshops, and researcher interviews.

Selected Awards

- 2025 — Two Huawei HQ Letters of Appreciation; High-Value Patent Recognition for EPD Disaggregation.
- 2024 — Director-on-the-Spot Award (top 5 in Vancouver R&D); Future Star Award (top 2 in the team).
- 2023- — Canada R&D Outstanding Team Award (2023), Huawei Gold Medal Team Award.(2021)
- Earlier — KVPY Fellowship (40 selected across India); Accenture Innovation Jockey (1st of 1,000+ entries).

Education

University of British Columbia – MSc (Thesis), Computer Science

Vancouver, 2019

Research focus: multimodal learning, video question answering, probabilistic machine learning

Thapar Institute of Engineering & Technology – BE, Computer Science

Patiala, 2013